KAVYA AGRAWAL

PHD SCHOLAR

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SUMMARY

Geospatial data science researcher working at the interface of cryosphere-influenced hydrology, Earth system science, and groundwater systems, with a strong focus on machine learning, deep learning, artificial intelligence, remote sensing, and GIS-based spatial modelling. Experienced in integrating multi-source, multi-scale cryospheric and Earth system datasets, including satellite-derived surface temperature, elevation-dependent climate variables, and terrestrial water storage products, with hydrochemical observations. Currently pursuing a PhD with research focused on groundwater quality versus quantity, contamination vulnerability assessment, and the role of cryospheric processes such as snow accumulation, melt, and glacier-fed hydrology in regulating surface–subsurface water availability, with emphasis on model robustness, generalizability, and decision-relevant outcomes.

EDUCATION

2024 - Present	**TERI SCHOOL OF ADVANCED STUDIES, DELHI** PhD Natural & Applied Sciences	Coursework CGPA: 9
2021-2023	**DR. B.R. AMBEDKAR UNIVERSITY, AGRA** MA Geography	CGPA: 7.60
2018-2021	**UNIVERSITY OF DELHI, DELHI** BA (Honors) Geography	CGPA: 7.86
2016- 2018	**DELHI PUBLIC SCHOOL, ALIGARH** Senior Secondary School	CGPA: 9.80
2014- 2016	**ST. FRANCIS INTER COLLEGE, HATHRAS** Secondary School	CGPA: 9.20

EXPERIENCE

01/ 2026 - Present	Comparing and Benchmarking Deep Learning Models for Predicting Groundwater Contamination caused by Fluoride with the effects of Spatial Resolution on Prediction	Research Paper (In Progress)
07/ 2025 - 12 / 2025	Course: Principles of Remote Sensing (NRG169), Practical Learning Hours: 32, Course Type: Core (Semester 1)	Course Taught (Msc Geoinformatics)
09/ 2025 - 12 /2025	Predictive Modelling and Spatial Risk Mapping of Uranium Contaminated Groundwater in India using Automated Machine Learning	Research Paper (Concluded)
06/ 2025 - 10 /2025	Distribution, Speciation and Controls on As Mobilization in the Groundwater in Aquifers of the Middle Gangetic Plain, India	Conference Poster Presentation
02/ 2023 - 08/ 2023	Evaluating Spatial and Elevation-wise Daytime/Nighttime LST Trends Across the Indus River Basin	Research Paper (Published)

POST- GRADUATION PROJECTS

08/ 2021 - 10/ 2023

- Measuring the Problems and Emotions of Urband-Ward Migrants: A Case Study of Construction Workers in Delhi
- Monitoring the Effects of Light Pollution & Evaluating the Effectiveness of Indian Regulations
- Community awareness, participation, & CCDRR: Reflections from North-West India
- Experience of Migrant College Students During COVID-19 Pandemic
- Analyzing the Possibilities and Prospects of 24/7 Water Supply in Delhi, India.
- Geography of Ramayana

GRADUATION PROJECTS

09/ 2018 - 04/ 2021

- Forest Monitoring & Urban Sprawl Analysis Using GIS in Delhi
- LULC Change detection for 20 years in south-west Delhi
- Social Indicators & Status Analysis of Jammu & Kashmir
- Socio-Economic Impact of Covid 19 in Delhi

PUBLICATIONS

November, 2023 — Mal, S., Agrawal, K., Rani, S. et al. Evaluating spatial and elevation-wise daytime/nighttime LST trends across the Indus River Basin. J. Mt. Sci. 20, 3154–3172 (2023). https://doi.org/10.1007/s11629-023-8157-8 Link to the paper

SKILLS

- GIS & RS Methods & Techniques
- ArcGIS, QGIS, Google Earth
- Google Earth Engine, ERDAS
- Disaster risk reduction and resilience concepts
- Socio-environmental data analysis
- Academic writing and conference presentations
- Spatial Data Analysis & Visualization
- Python, JavaScript
- Climate–water–environment interactions
- Sustainability and climate policy exposure
- Field-based and survey-based research
- Rapporteurship and policy session documentation
- Groundwater contamination assessment (drinking water quality)
- Scientific communication and reporting
- Editorial and content development
- Academic coordination and event support
- Interdisciplinary collaboration

CONFERENCES & PAPER PRESENTATIONS

- Presented a research paper titled “Distribution, Speciation and Controls on As Mobilization in the Groundwater in Aquifers of the Middle Gangetic Plain, India” at the 9th International Congress & Exhibition on Arsenic in the Environment (As2024), themed Arsenic and Other Pollutants, Water Security and One Health under Global Climate Change Scenario, KIIT Deemed to be University, Bhubaneswar, Odisha, October 20–24, 2024.
- Presented a research paper titled “Migrant College Students during Covid Pandemic: Evidence from the Field” at an International Conference on Challenges to Disaster Risk Reduction and Resilient Habitat, organized by Centre for Disaster Management Studies, Shaheed Bhagat Singh College, University of Delhi, April 5–6, 2022.
- Presented a research paper titled “Migrant College Students during Covid Pandemic: Evidence from the Field” at the Jamia International Conference on Education-2022 on Outcome-Based Curriculum and Pedagogical Demands in the Post-Covid Era, organized by Department of Teachers Training & Non-Formal Education (IASE), Faculty of Education, Jamia Millia Islamia, New Delhi, May 6–8, 2022.
- Presented a research paper titled “Digital Enhancement of Physical Space and Connective Turn in Tourism Industry during and Post-COVID-19 Period” at an International Conference on Geography and the Digital: Intersections and Conflations, organized by the Department of Geography, Delhi School of Economics, University of Delhi, May 23–25, 2022.
- Presented a research paper titled “Educational Relevance of Ramayana in Modern India” at a National Seminar on Promoting Human Values and Culture for Peaceful Co-Existence, organized by Amity Centre for Sanskrit and Indic Studies, Amity School of Liberal Arts and Amity Academic Staff College, Amity University Haryana, sponsored by ICPR, July 22, 2022.
- Presented a research paper titled “Problems and Prospects of Delhi’s Drinking Water Supply” at the IGU Thematic Conference on Sustainability, Future Earth and Humanities: Opportunities and Challenges, organized by the Department of Geography, Central University of Haryana, Mahendragarh, November 24–25, 2022.

TRAINING, CAPACITY BUILDING & SCHOOLS ATTENDED

- One-week Capacity Building Program on Harnessing Geospatial Technology for Forest Fire Disaster Management: Mapping, Monitoring and Mitigation, Amity University, Noida, March 11–15, 2024.
- 21-day Winter School Capacity Building Program in Geospatial Science and Technology, supported by National Geospatial Program (DST, Government of India), Amity University, Noida, November 18 – December 7, 2024 (Offline).
- Capacity Building Workshop on UAV/Drone Applications in Agriculture and Natural Resource Management, TERI SAS in collaboration with IIT BHU, IIT Roorkee, Namami Gange, Hokkaido University, GLP Earth, GLP Japan, Abhipsa Foundation, and Kambil, December 10–12, 2024.
- QGreenland Educator Workshop (Virtual), focused on developing educational activities using QGreenland data, August 5–7, 2025.

SUMMITS & POLICY EVENTS

- World Sustainable Development Summit (WSDS) 2025, themed Partnerships for Accelerating Sustainable Development and Climate Solutions, New Delhi, March 5–7, 2025.
- Science Summit 2025, Session: Marine Resources for Sustainable Development, September 16, 2025.
- Act4Earth Dialogue (Virtual) titled Strengthening Multilateralism on the Road to COP30 and Beyond, October 17, 2025.

RAPPORTEUR / ACADEMIC & POLICY ENGAGEMENT

- Rapporteur, Ministerial Session: NDC 3.0 and Multilateralism on the Road to COP30 and COP33, World Sustainable Development Summit 2025, TERI.
- Rapporteur, High-Level Plenary on Achieving Sustainable Development through Integrated Action, World Sustainable Development Summit 2025, TERI.

EXPERT TALKS, SEMINARS & WORKSHOPS ATTENDED

- Lecture by Professor John A. Cherry, Lee Kuan Yew Water Prize recipient, on groundwater and sustainable development, October 24, 2024.
- Expert talk on DHI Urban Water Solutions by Mr. Morten Just Kjølby (Product Portfolio Manager, MIKE+, DHI Denmark), organized by Department of Natural and Applied Sciences, TERI School of Advanced Studies, October 7, 2025.
- Roots to Resilience Seminar, organized by TERI SAS in collaboration with UNEP, UNESCO, and Akashvani, July 25, 2025.
- Workshop on Climate Skills: Seeds for Transition, organized by TERI SAS in collaboration with British Council and HSBC Bank.

MISCELLANEOUS

POSITIONS OF RESPONSIBILITY & SERVICE

- Judge / Panel Member, Sustainability Hackathon 2025, TERI SAS (School Category). NSS Volunteer, 2019–2021
- Editor-in-Chief, Geography Association, University of Delhi, 2018–2021
- Magazine Coordinator, Vani, SBSC, University of Delhi, 2018–2020
- English Editor, Vani, SBSC, University of Delhi, 2019
- Volunteer, SWAPO NGO, 2016–2023
- Volunteer Coordination, Vasundhara, 2018–2019

AWARDS & ACHIEVEMENTS

- Consecutive Top 3 Ranks in English Olympiads, British Olympiad Foundation, 2015–2018
- Merit Awards, 2016, 2018
- State Child Scientist Award, National Science Congress, 2018
- 20 Cultural and General Knowledge Awards from various organizations
- 2nd Rank in Map making competition organized by Amity University 2025.

COVER LETTER

Name: Kavya Agrawal

University Affiliation: TERI School of Advanced Studies, Vasant Kunj, New Delhi, India

Position: Doctoral Researcher

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This is my formal application for the GlaMacLeS Summer Course on Machine Learning for Glaciology to be held from July 4–13, 2026.

I am currently a doctoral researcher at TERI SAS, working on groundwater contamination and quality/quantity assessments using machine learning, deep learning, artificial intelligence, and remote sensing based Earth system datasets.

My doctoral research integrates hydrogeochemical observations with spatially continuous environmental predictors derived from satellite remote sensing, land surface models, and reanalysis products. I work extensively with large-scale datasets such as terrestrial water storage anomalies, precipitation, land surface temperature, soil moisture, and vegetation indices, and apply machine learning and deep learning approaches for spatial prediction, benchmarking, and interpretability. This work has provided me with a strong methodological foundation in data-driven environmental modelling.

Although my primary research focus is groundwater systems, my work intersects with broader Earth system processes that influence terrestrial water storage and hydrological variability at regional to continental scales, including differences observed between basin-based and mountain-based systems. While groundwater contamination has been extensively studied in surface and subsurface systems, cryospheric–groundwater interactions remain comparatively underexplored and are governed by distinct physical processes.

I am therefore interested in extending my research to better understand groundwater contamination under cryospheric-influenced conditions to create moderate intersections for studying holistic groundwater based nitty-gritties that apply to heterogeneous conditions for strengthening the physical grounding and transferability of my research.

Participation in the GlaMacLeS course would be beneficial to my research and professional development by providing focused training in glaciology-oriented machine learning approaches, including deep learning for cryospheric remote sensing. The interactive lectures and project-based structure of the course in glaciology would help me responsibly integrate these methods into my existing research framework and broaden my interdisciplinary understanding of groundwater irrespective of geographic phenomena, combining machine learning, glaciology and hydrogeology.

DESCRIPTION OF PROPOSED THESIS / PRIMARY RESEARCH INTERESTS

My primary research interests lie at the interface of Earth system science, remote sensing, and machine learning, with focus on cryosphere-influenced hydrological processes. While my broader doctoral work is centered on large-scale water system analysis, I am particularly interested in understanding how glaciological and cryospheric processes regulate terrestrial water storage and hydrological variability across regional to continental scales.

Cryospheric processes such as snow accumulation, seasonal melt, and glacier-fed hydrology play a critical role in controlling surface and subsurface water availability, especially in mountain and high-latitude systems. Differences observed between basin-based and mountain-based hydrological behavior highlight the distinct influence of cryosphere-dominated regimes, which are governed by processes that differ fundamentally from non-cryospheric environments. Despite their importance, cryosphere-influenced subsurface and groundwater-related processes remain comparatively underexplored, particularly from a data-driven and machine learning perspective.

My research interests therefore increasingly emphasize the need to incorporate cryospheric controls into large-scale environmental modeling frameworks. Satellite-derived observations and Earth system datasets inherently integrate cryospheric signals, making them well suited for machine learning–based analysis. However, leveraging these data effectively requires a strong understanding of glaciology-specific modeling challenges.

Machine learning provides a powerful set of tools for addressing these challenges, particularly through approaches such as deep learning for cryospheric remote sensing, adjoint-based optimization and backpropagation, and model emulation of complex physical systems. I intend to develop a formal and structured understanding of these methods as applied specifically to glaciological systems by ensuring that data-driven models remain physically grounded, interpretable, and transferable across cryosphere-dominated environments.

Participation in the GlaMacLeS Summer Course would directly support this transition by providing focused training in machine learning approaches developed within the glaciology community. Exposure to glaciology-oriented ML frameworks, combined with project-based learning and interaction with researchers actively working in cryospheric science, would enable me to understand and to responsibly integrate these methods into my research trajectory and contribute to interdisciplinary work at the intersection of glaciology, hydrology, and machine learning.

Major collaborations included as part of this effort involve the use of national and global Earth system datasets produced by space agencies and research institutions, as well as academic supervision at my home institution. These data-driven collaborations provide the foundation for extending my work toward cryosphere-focused applications including the guidance and skill received at the GlaMacLeS Summer Course on Machine Learning for Glaciology.